

Battery Datasheet - G2_kappa0175

****Case**:** t6_r1_singlemodel | ****Document**:** VBF-T6R1SINGLEMODEL-DSH-001 | ****Model**:** SPMe (the DFN cross-check noted)

Electrical

Parameter	Value	Condition
Nominal capacity	5.4142 Ah	1C discharge to 2.5 V
Discharge midpoint voltage	4.2078 V	1C
Volumetric energy density	1107.02 Wh/L	the calc-energy output
Gravimetric energy density	477.18 Wh/kg	the calc-energy output
Cell mass / volume	47.4 g / 20.44 cm3	
Thickness	199 um	the stack
DC resistance	5.39307e-07 ohm	the calc-energy output

Fast charge (the 4C protocol: the 1C discharge to 2.5 V, then the 4C charge to 4.7 V, the 45 degC ambient)

Parameter	Value	Note
4C charge duration	64.7 s	to the 4.7 V voltage limit
4C charge capacity	0.0719 Ah (1.33% of the nominal)	voltage-limited
Lithium plating	NONE (the anode surface potential min +0.0454 V)	the whole protocol
Max temperature	321.442 K	<= 323.15 K (50 degC) PASS

Aging (the 1C x 100 cycles at the 45 degC, the SEI growth)

Parameter	Value
SEI thickness after 100 cycles	133.56 nm (<= 500 nm PASS)

Verification note

DFN cross-check: model DFN, T_max 321.238 K, anode surface potential min 0.0071 V (plated: False), capacity metric 0.3954 Ah.

Honest limitations

The 4C fast-charge profile is voltage-limited: the charge terminates after ~65 s at the 4.7 V limit with ~1.3% of the capacity delivered. The no-plating margin is 45 mV on the anode surface potential. The electrolyte conductivity 0.175 S/m is an extreme low-salt/quasi-solid formulation point.